

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE:** DSA04

COURSE OUTCOME COVERED

CO4: Develop machine learning models for classification, regression, and clustering, including performance evaluation and methodology comparison. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 10 Total Marks:100 Annexure A

VIVA VOCE

Code of Conduct:

*I **Anto Chriswin G 192324093** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

SCENARIO BASED ASSESSMENT							
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Level	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		

		and factors involved					
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Annexure A

1. How does Linear Regression works?

- Linear Regression finds the **relationship between input and output variables**.
- It fits a **straight line** through the data points.
- It predicts the dependent variable using the independent variable(s).

2. What is the objective of Linear Regression?

- The main objective is to predict a continuous numerical value.
- It tries to find the best-fitting line for the given data.
- It minimizes the difference between actual values and predicted values, usually using Mean Squared Error (MSE).

3. What is the difference between dependent and independent variables?

- Independent variable (X) is the input or predictor variable.
- Dependent variable (Y) is the output that we want to predict.
- Example: If we predict house price using house size, house size is independent and house price is dependent.

4. Explain the equation of a simple linear regression model.

- The equation is $Y = \beta_0 + \beta_1 X$.
- Y is the predicted dependent variable, and X is the independent variable.
- β_0 is the intercept and β_1 is the slope/coefficient.

5. What does the intercept represent in a regression equation?

- The intercept is the value of Y when X is zero.
- It represents the starting or baseline value of the dependent variable.
- In $Y = \beta_0 + \beta_1 X$, β_0 represents the intercept.

6. What does the coefficient (slope) indicate?

- The coefficient tells us how much Y changes when X increases by one unit.
- A positive coefficient means Y increases as X increases.
- A negative coefficient means Y decreases as X increases.

7. What is the meaning of R-squared (R^2)?

- R^2 measures how well the model explains the variation in the dependent variable.
- Its value generally ranges from 0 to 1.
- For example, $R^2 = 0.80$ means the model explains approximately 80% of the variation in the dependent variable.

8. What assumptions are made in Linear Regression?

- There should be a linear relationship between the independent and dependent variables.
- The observations should be independent of each other.
- The residuals should have constant variance (homoscedasticity) and should be approximately normally distributed for standard statistical inference.

9. What is the difference between Simple and Multiple Linear Regression?

- Simple Linear Regression uses one independent variable.
- Multiple Linear Regression uses two or more independent variables.
- Example: $\text{Price} = \beta_0 + \beta_1(\text{Size})$ is simple, while $\text{Price} = \beta_0 + \beta_1(\text{Size}) + \beta_2(\text{Bedrooms})$ is multiple.

10. How do you evaluate the performance of a Linear Regression model?

- R^2 measures how much variation is explained by the model.
- MAE (Mean Absolute Error) measures the average absolute prediction error.
- MSE and RMSE measure prediction errors, where lower values generally indicate better performance.

11. What is Logistic Regression?

- Logistic Regression is a supervised machine learning algorithm.
- It is mainly used for classification problems.
- It predicts the probability of a class, such as 0/1 or Yes/No.

12. Why is Logistic Regression used for classification problems?

- It converts the output into a probability between 0 and 1.
- A threshold, commonly 0.5, can be used to assign a class.

- It works well for binary classification problems such as spam/not spam or churn/not churn.

13. What is the sigmoid (logistic) function?

- The sigmoid function converts any numerical value into a value between 0 and 1.
- Its formula is $\sigma(z) = 1 / (1 + e^{-z})$.
- It is used in Logistic Regression to calculate the probability of the positive class.

14. What is the range of output values in Logistic Regression?

- The output probability ranges from 0 to 1.
- A value close to 0 means a low probability of the positive class.
- A value close to 1 means a high probability of the positive class.

15. What is the difference between Linear and Logistic Regression?

- Linear Regression is mainly used to predict continuous numerical values.
- Logistic Regression is mainly used for classification.
- Linear Regression produces continuous output, while Logistic Regression produces a probability between 0 and 1.

16. What is a decision boundary?

- A decision boundary is a boundary that separates different classes.
- It determines which class a new data point belongs to.
- In binary Logistic Regression, a common decision threshold is 0.5.

17. How is accuracy measured in Logistic Regression?

- Accuracy is the proportion of correct predictions out of all predictions.
- Formula: $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$.
- For example, if 90 out of 100 predictions are correct, the accuracy is 90%.

18. What are True Positive, False Positive, True Negative, and False Negative?

- True Positive (TP): Model correctly predicts the positive class.
- False Positive (FP): Model predicts positive, but the actual class is negative.
- True Negative (TN): Model correctly predicts the negative class.
- False Negative (FN): Model predicts negative, but the actual class is positive.

19. What is Ridge Regression?

- Ridge Regression is a regularized version of Linear Regression.
- It adds an L2 regularization penalty to the loss function.
- It reduces the size of model coefficients to help prevent overfitting.

20. Why is Ridge Regression used?

- It is used to reduce overfitting in regression models.
- It is useful when independent variables are highly correlated with each other.
- It keeps all features in the model but shrinks their coefficients toward zero.

21. What is regularization?

- Regularization is a technique used to reduce overfitting in machine learning models.
- It adds a penalty term to the model's loss function.
- It controls the size of model coefficients to make the model more generalizable.

22. What is the role of the regularization parameter (λ) in Ridge Regression?

- λ controls the strength of regularization.
- A larger λ causes the coefficients to become smaller.
- If λ is too large, the model may become too simple and underfit.

23. Which type of penalty is used in Ridge Regression?

- Ridge Regression uses an L2 penalty.
- The penalty is based on the squared values of the coefficients.
- The penalty term is commonly written as $\lambda \sum \beta^2$.

24. How does Ridge Regression handle multicollinearity?

- Ridge reduces the size of coefficients when predictors are highly correlated.
- This makes the model more stable.
- It helps reduce the effect of multicollinearity on model predictions.

25. Can Ridge Regression reduce coefficients to exactly zero?

- Generally, No, Ridge does not reduce coefficients exactly to zero.
- It shrinks coefficients toward zero.
- Therefore, Ridge usually keeps all features in the model.

26. What is the difference between Linear Regression and Ridge Regression?

- Linear Regression does not use regularization.
- Ridge Regression adds an L2 penalty to the loss function.
- Ridge is useful for reducing overfitting and multicollinearity.

27. What is Lasso Regression?

- Lasso stands for Least Absolute Shrinkage and Selection Operator.
- It is a regularized version of Linear Regression.
- It uses regularization to reduce coefficients and perform feature selection.

28. Which type of regularization is used in Lasso Regression?

- Lasso Regression uses L1 regularization.
- It applies a penalty based on the absolute values of coefficients.
- The penalty encourages some coefficients to become exactly zero.

29. What is the mathematical penalty term used in Lasso?

- The Lasso penalty is based on the sum of absolute coefficient values.
- It is written as $\lambda \sum |\beta|$.
- λ controls the strength of the L1 penalty.

30. How does Lasso perform feature selection?

- Lasso can shrink some coefficients to exactly zero.
- Features with zero coefficients are effectively removed from the model.
- Therefore, Lasso can automatically identify and select important features.

31. What happens when λ increases in Lasso Regression?

- The regularization becomes stronger.
- More coefficients may shrink toward or become zero.
- If λ is too large, important features may also be removed, causing underfitting.

32. What is the difference between Ridge and Lasso Regression?

- Ridge uses L2 regularization, while Lasso uses L1 regularization.
- Ridge usually shrinks coefficients close to zero, while Lasso can make them exactly zero.
- Therefore, Lasso can perform feature selection, while Ridge generally does not.

33. Why is Lasso useful in high-dimensional datasets?

- High-dimensional datasets contain many features.
- Lasso can remove less important features by making their coefficients zero.
- This produces a simpler and more interpretable model.

34. Can Lasso eliminate irrelevant features completely?

- Yes, Lasso can reduce irrelevant feature coefficients to exactly zero.
- A coefficient of zero means that feature does not contribute to the model's prediction.
- This makes Lasso useful for automatic feature selection.

35. What is K-Nearest Neighbors (KNN)?

- KNN is a supervised machine learning algorithm.
- It can be used for classification and regression.
- It predicts an output based on the nearest data points.

36. Is KNN a supervised or unsupervised learning algorithm?

- KNN is a supervised learning algorithm.
- It learns from labeled training data.
- The labels of nearby data points are used to make predictions.

37. How does KNN classify a new data point?

- KNN calculates the distance between the new point and training points.
- It selects the K nearest neighbors.
- The class with the majority vote among those neighbors becomes the predicted class.

38. What does the value of K represent?

- K represents the number of nearest neighbors considered for prediction.
- For example, $K = 5$ means the model considers the 5 nearest data points.
- The value of K can significantly affect the model's performance.

39. What happens if K is too small?

- The model becomes sensitive to noise and outliers.
- It may overfit the training data.
- Predictions can change significantly because of individual nearby points.

40. What happens if K is too large?

- The model considers too many neighboring points.
- It may ignore local patterns and underfit the data.
- The decision boundary becomes smoother and may reduce classification accuracy.

41. Which distance measures are commonly used in KNN?

- Euclidean distance is the most commonly used distance measure.
- Manhattan distance measures the distance by summing the absolute differences.
- Minkowski distance is a general form that includes Euclidean and Manhattan distance.

42. Why is feature scaling important in KNN?

- KNN uses distance to find the nearest neighbors.
- Features with larger numerical values can dominate the distance calculation.
- Scaling puts features on a similar range and helps KNN make fairer predictions.

43. What is a Decision Tree?

- A Decision Tree is a supervised machine learning algorithm.
- It can be used for classification and regression.
- It makes decisions by splitting data into smaller groups based on feature conditions.

44. What are the main components of a Decision Tree?

- Root node: The first and top-level decision point.
- Branches: Represent the outcomes of a decision or condition.

- Leaf nodes: Represent the final prediction or output.

45. What is the root node in a Decision Tree?

- The root node is the topmost node of the tree.
- It represents the first feature used to split the data.
- It is usually selected because it provides the best split according to a criterion such as information gain or Gini impurity.

46. What is a leaf node?

- A leaf node is the final node in a Decision Tree.
- It does not have any further branches.
- It represents the final prediction, such as a class or numerical value.

47. What is entropy in Decision Trees?

- Entropy measures the impurity or uncertainty in a dataset.
- Low entropy means the data is mostly from one class.
- High entropy means the data contains a more equal mixture of different classes.

48. What is information gain?

- Information Gain measures how much uncertainty is reduced after a data split.
- It is calculated using the difference between the original entropy and the weighted entropy after splitting.
- A Decision Tree generally chooses the split with the highest information gain.

49. What is overfitting in Decision Trees?

- Overfitting occurs when the tree becomes too complex and learns the training data too closely.
- It may perform very well on training data but poorly on unseen data.
- Very deep trees are more likely to overfit.

50. How can pruning improve a Decision Tree model?

- Pruning removes unnecessary branches or nodes from the tree.
- It reduces the complexity of the model.
- It helps reduce overfitting and can improve performance on unseen data.